RiskLens

Investor Risk Assistant

Decision-support tool for quantitative financial asset risk assessment

Monte Carlo	GARCH	Markov-Switching
EVT	XGBoost	Backtesting

Agenda

01 **The Problem**
Why traffic-light risk ratings fall short

02 **What is RiskLens?**
Concept, data sources and architecture in 1 minute

03 **The 3 Model Tiers**
From simple to complex — and why it matters

04 **Monte Carlo for Non-Statisticians**
Rolling 10,000 dice instead of guessing

05 **Key Assumptions**
What the models assume (and where they can break)

06 **Findings & Conclusions**
What we discovered when we validated the models

07 **Demo / Q&A**
Interactive Streamlit dashboard

CONCEPT

Why Predict Risk, Not Price?

Price → not predictable (noise + non-stationary)

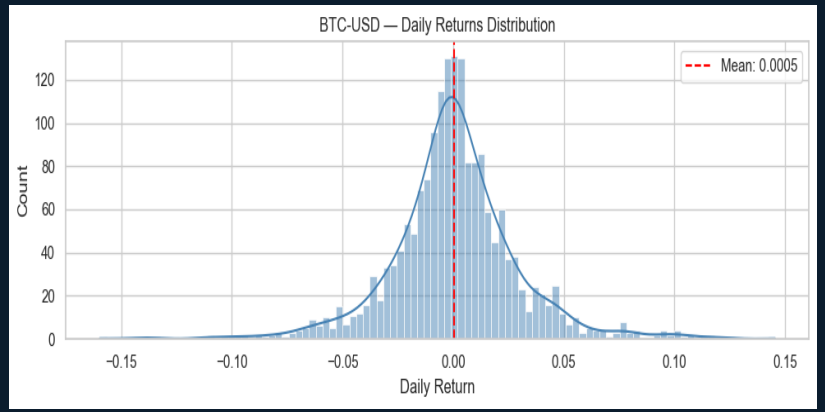
Risk → estimable (return distribution)

The model does not predict the exact point.

It evaluates: probable deviation and extreme events.

We model stochastic chaos, not deterministic paths

💡 decisions = loss function, not trajectory



How Much Can I Actually Lose?

Investors need numbers, not colors. RiskLens translates risk into dollars and probabilities.

Traditional Analysis	RiskLens
• "High risk" / "Medium" / "Low"	• 80% prob. of losing between $1.2M and $3.8M
• Subjective 1-to-5 scales	• 3 models with increasing complexity
• No quantified potential losses	• Monte Carlo simulation: 10,000 scenarios
• Cannot compare assets objectively	• Statistical backtesting (Kupiec, Christoffersen)
• No way to know if your hedge works	• Each model tells you WHERE and WHY it fails

What is RiskLens?

A Python + Streamlit tool combining historical analytics, Monte Carlo simulation and statistical backtesting.

Data

Yahoo Finance (yfinance).
Daily log-returns.
252 trading days = 1 year.
Policy: 1 file per asset,
overwrite on refresh.

Models

3 simulation tiers
(Baseline → GARCH+t
→ MS-GARCH+EVT).
+ EVT and XGBoost
as independent auditors.

Validation

Rolling-window backtest.
Kupiec (coverage) +
Christoffersen (independence).
271 automated tests.
Per-regime comparison.

What Are the Main Parameters?

VaR (Value at Risk)

The percentile threshold.

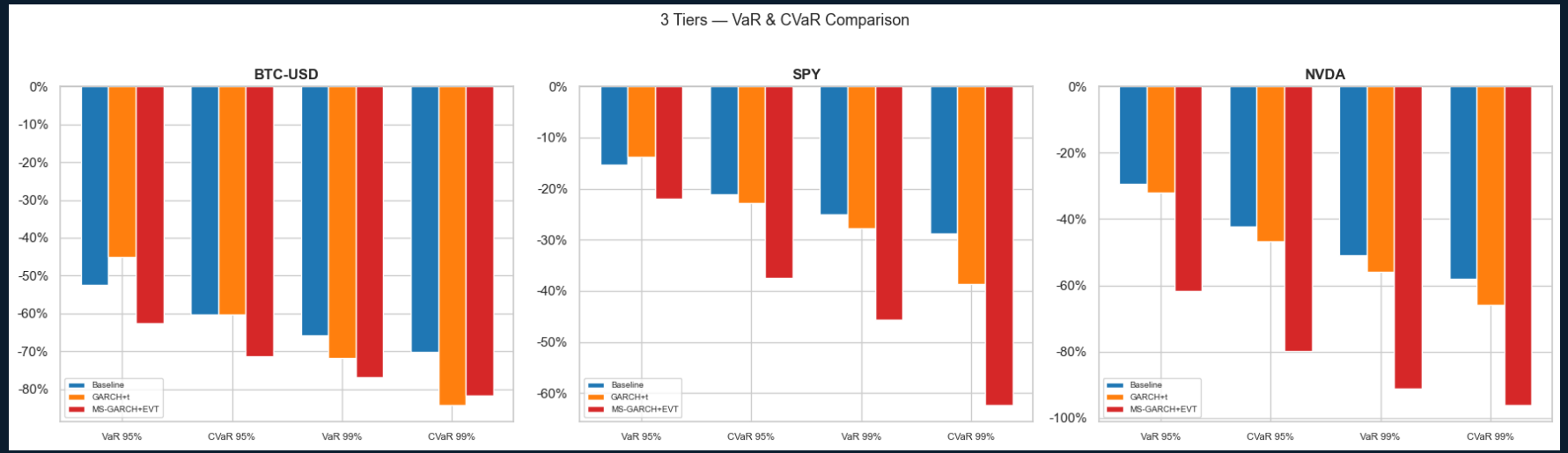
How far the boat typically deviates.

CVaR (Expected Shortfall)

The tail mean.

What happens if it hits the storm.

💡 VaR measures frequency, CVaR measures severity



The 3 Modeling Tiers

Each tier adds a layer of realism. But more complexity ≠ better. Each one fails differently.

TIER 1
Baseline

Constant volatility · Normal shocks

✓ Simple, stable, easy to understand

✗ Regime-blind. 0% breach in calm, 20.9% in crisis.

TIER 2
GARCH + Student-t

Dynamic vol (GARCH 1,1) · Fat-tailed shocks

✓ Captures volatility clustering

✗ Post-crisis amnesia: σ decays too fast. 34.9% breach in crisis.

TIER 3
MS-GARCH + EVT

GARCH + Markov regimes · Per-regime GPD tails

✓ Best calibrated (5.1% breach ≈ 5% expected)

✗ Detection lag at regime turns. Small crisis sample.

TIER 1

Baseline: Geometric Brownian Motion

Continuous random motion · Normal increments · Constant volatility

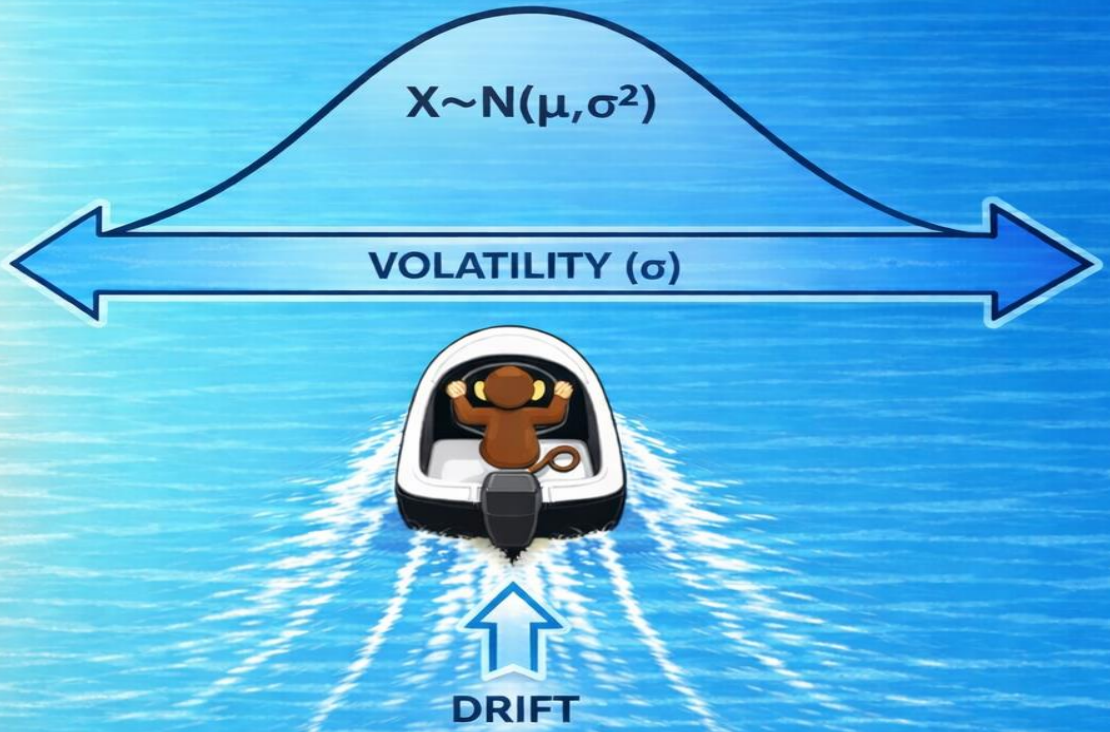
Limitation: Does not capture clustering or fat tails.

Metaphor: Sea is always the same. Constant noise.

💡 baseline = benchmark, not reality

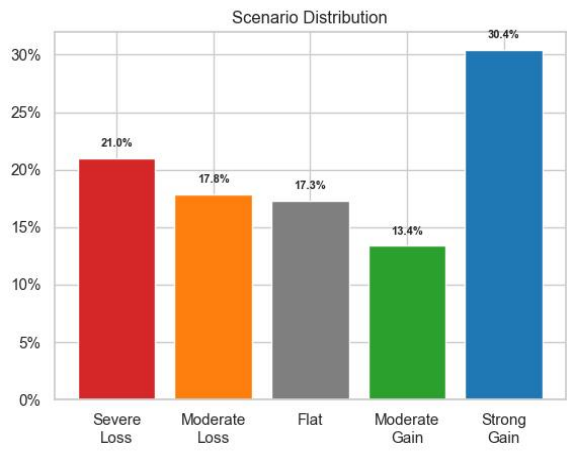
$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t$$

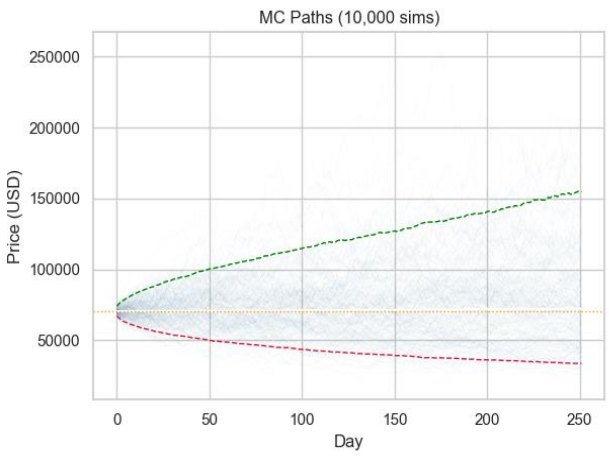
VAR
(Value at Risk)

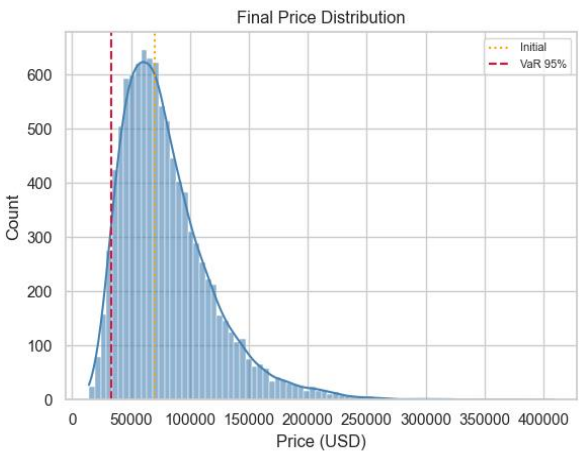


Monte Carlo for Non-Statisticians

Instead of predicting ONE future, we simulate 10,000 possible futures and see how they distribute.







VaR 95%

"With 95% confidence, you won't lose more than X%." The 5th percentile cut.

CVaR

"IF the loss exceeds VaR, how much do you lose on average?" Captures worst-case severity.

10,000 paths

Each path is a simulated price trajectory over N days, using the selected volatility model.

The Process in 4 Steps

1

Ingest Data

Download prices from Yahoo Finance. Compute daily log-returns. Clean NaN.

2

Fit Model

Choose tier. Fit GARCH, HMM, GPD per tier. Window: 252 days.

3

Simulate 10K Paths

Monte Carlo generates price scenarios. Extract VaR, CVaR, probabilities.

4

Validate

Rolling backtest. Kupiec: is breach rate correct? Christoffersen: are breaches independent?

Key Assumptions

Every model rests on assumptions. Knowing them is mandatory to interpret results correctly.

Daily log-returns

Additive over time, approximately symmetric. 252 trading days = 1 year. Annualization factor √252.

252-day fit window

Trade-off: long enough to fit GARCH stably, short enough not to blend structurally different regimes.

GARCH: $\alpha+\beta < 1$

Required for finite unconditional variance. Typical persistence ~0.96 implies a half-life of ~19 days.

No leverage effect

Positive and negative shocks of same size produce equal future vol. Bias: underestimates post-crash vol.

2 regimes (fixed HMM)

Calm and crisis. No BIC/AIC search. Sufficient for the bimodality observed in equities and crypto.

GPD: iid exceedances

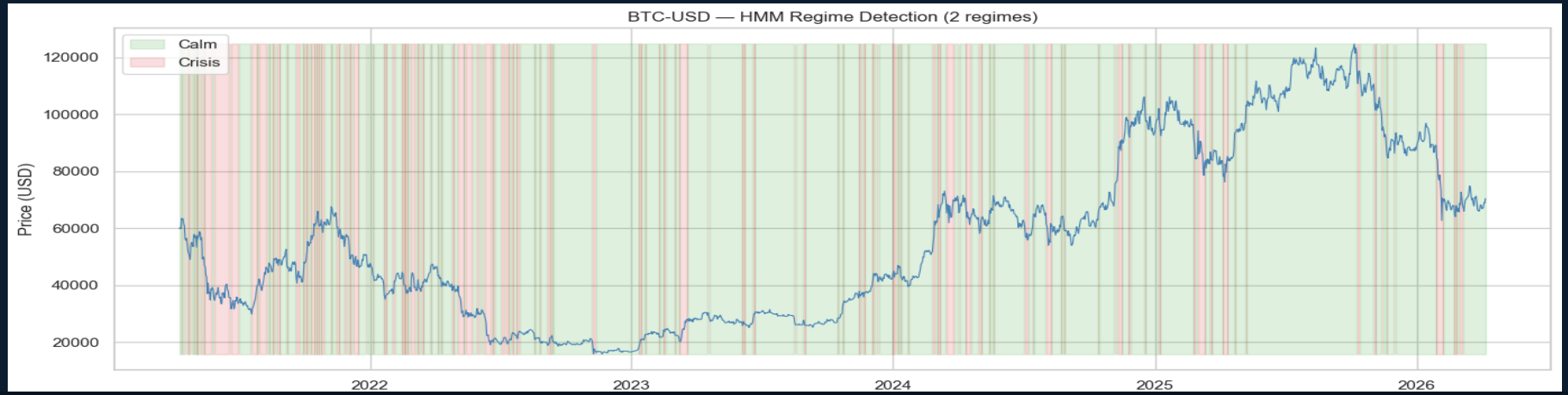
Peaks over threshold treated as independent. In reality they cluster during volatility episodes.

Time-Series Structure/Approach

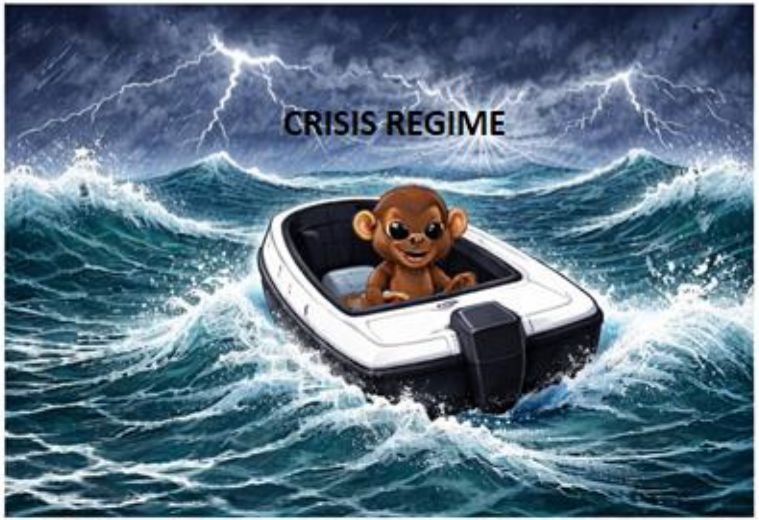
Continuous-time volatility structure : Rolling volatility → σ(t) + Discrete regime switching (latent states)

2 Regimes
Calm (calm sea) → stable behavior
Crisis (storm) → different dynamics.

💡 there is no single generating process



CALM REGIME


CRISIS REGIME

DO NOT BE
THAT MONKEY

STUDY YOUR RISK
WITH RISKLENS

TIER 2

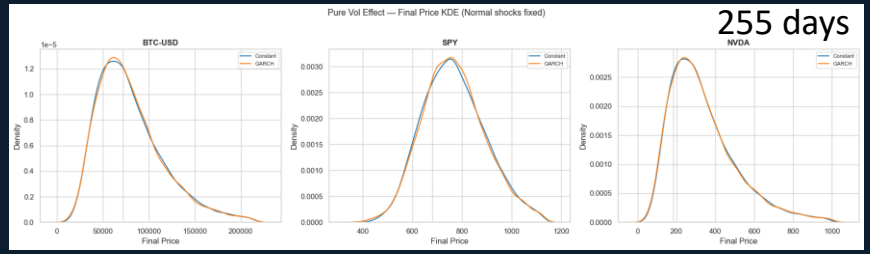
Volatility Upgrade: GARCH

Volatility depends on the past · Volatility clustering

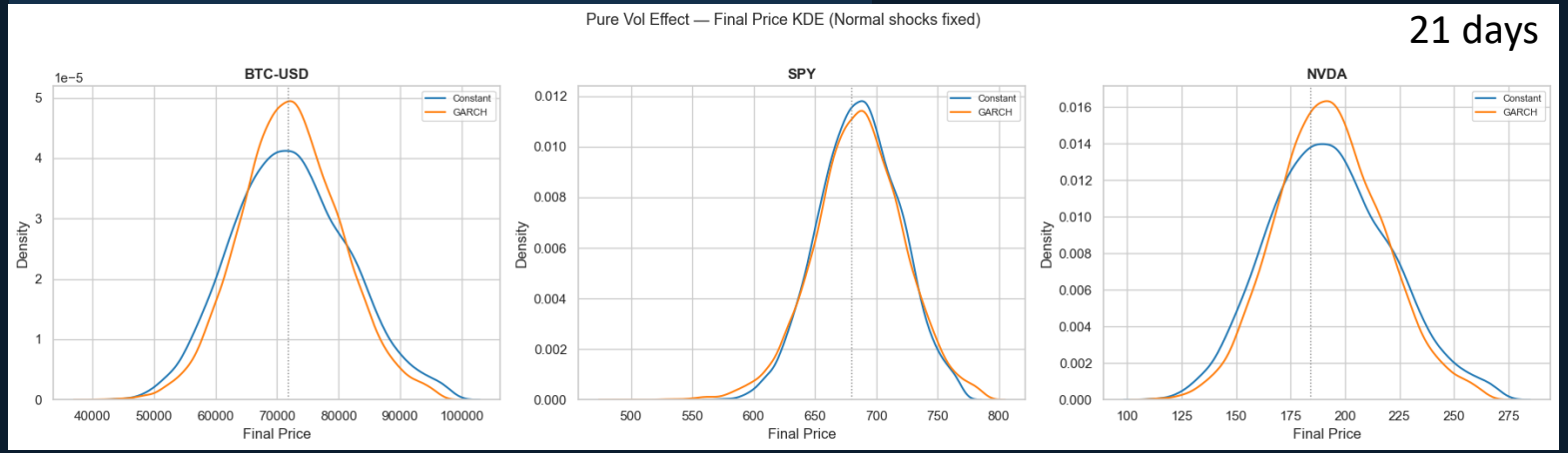
GARCH(1,1): $\sigma^2(t)$ = function of past shocks + past variance

Volatility impact diminishes over longer horizons due to averaging effects

Metaphor: If there was a storm → waves persist.



💡 volatility has memory



Distribution: Study of Tails

Normal

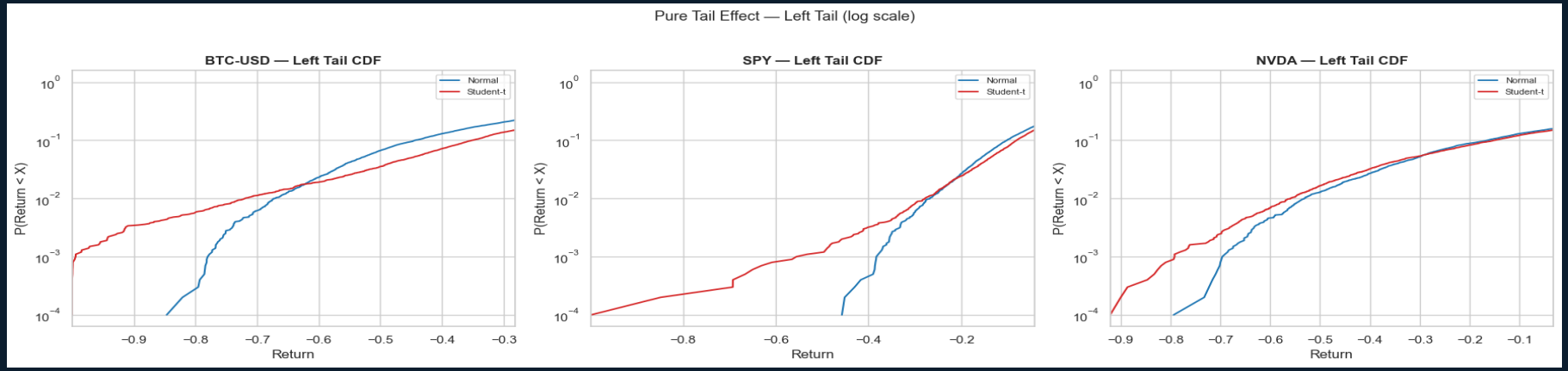
Simple. Thin tails.

Student-t

Fat tails. More realistic.

EVT

Models extremes directly.

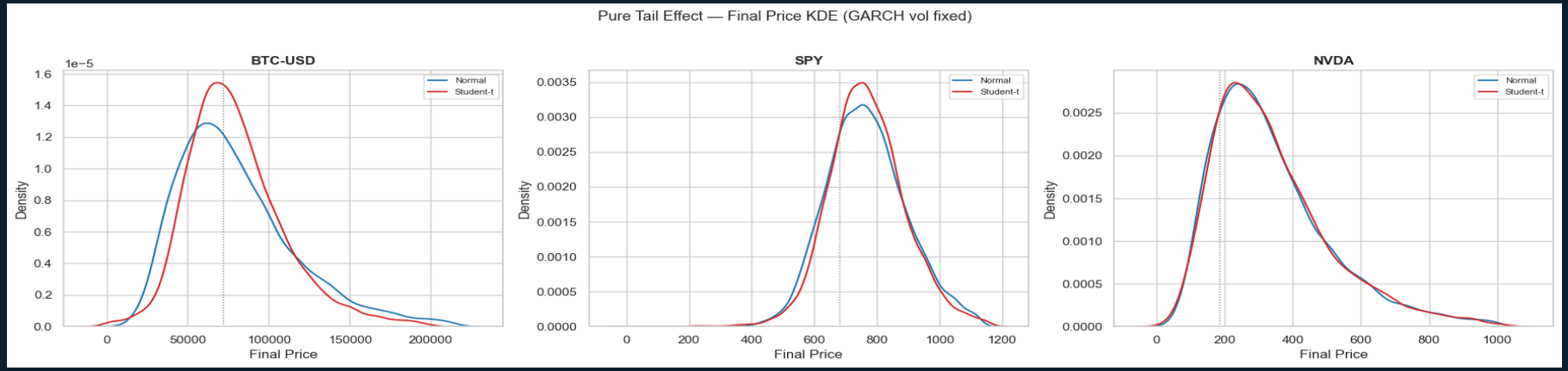


Tails Insight

Distribution choice = direct impact on VaR/CVaR

Normal → falsely calm sea | Student-t → large waves | EVT → real storms

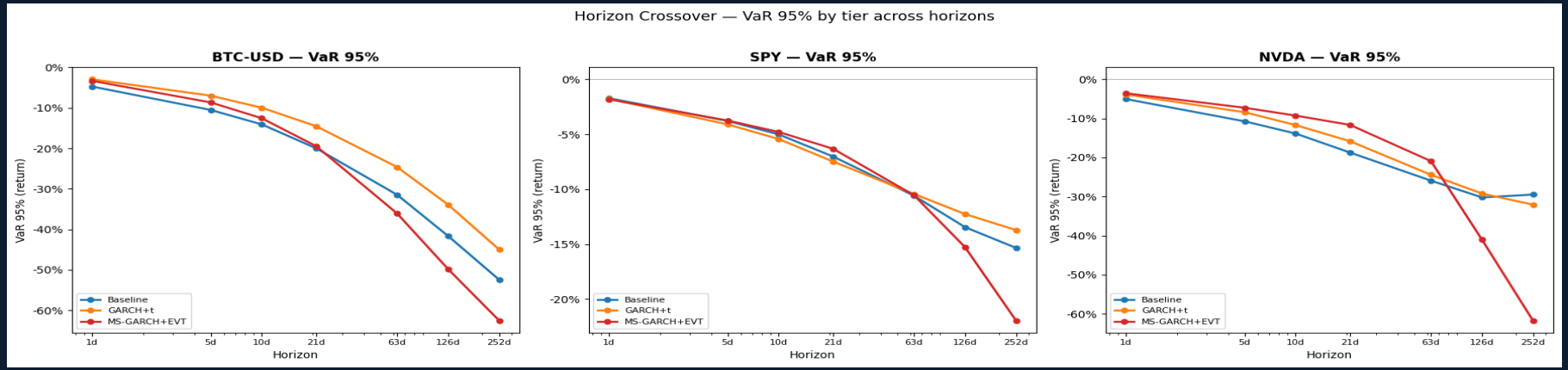
💡 risk lives in the left tail



Finding #1: Horizon Changes Everything

There is no single "best model." There is a best model for a given horizon on a given asset.

Asset	Crossover	Interpretation
SPY	≈ 5 days	MS-GARCH+EVT overtakes GARCH+t almost immediately
BTC-USD	≈ 63 days	GARCH+t wins short-term; EVT dominates after ~2 months
NVDA	≈ 126 days	Takes half a year for EVT to add value over GARCH

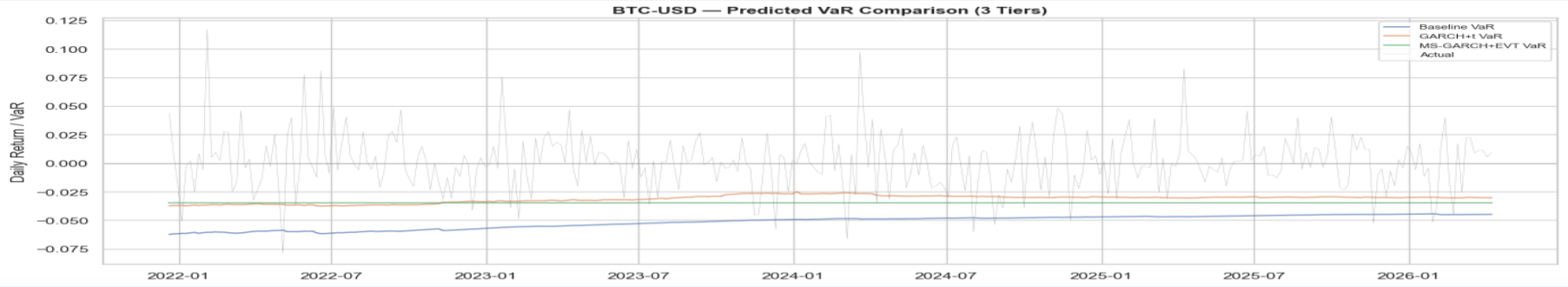


Finding #2: The Backtest as Judge

Rolling-window VaR 95% on BTC-USD. n=2,000 sims/window, window=252d, step=5d, seed=42.

Tier	Breach rate	Kupiec	Christoffersen	Verdict
Baseline	2.5%	**FAIL (p=0.027)**	PASS	Too conservative
GARCH+t	6.0%	PASS (p=0.415)	PASS	Passes, but 34.9% breach in crisis
MS-GARCH+EVT	**4.1%**	**PASS (p=0.464)**	**PASS (p=0.554)**	**Best calibrated ✓**

Expected: 5% breaches. MS-GARCH+EVT = 4.1% (closest). Only tier passing both tests cleanly.



Finding #3: Each Tier Fails Differently

The headline finding. The "best" model depends on which failure mode you can tolerate.

	Calm Breach (n=272)	Crisis Breach (n=43)	Overall	Failure Mode
Baseline	0.0%	**20.9%**	2.9%	*Lagged level adjustment*
GARCH+t	3.7%	**34.9%**	7.9%	*Post-crisis amnesia*
MS-GARCH+EVT	1.1%	**30.2%**	5.1%	*Detection lag + small sample*

Baseline: "always answering the previous regime's question"

0% breach in calm (wastes capital) and 20.9% in crisis (dangerously under-reactive). The two errors partially cancel in the aggregate.

GARCH+t: "σ recovers too fast after crisis subsides"

The 19-day half-life is shorter than real vol clusters. σ is already back at base level when the next shock arrives.

MS-GARCH+EVT: "the HMM needs time to confirm the switch"

The failure is structural (detection lag), not parametric. The only tier where the remaining error is from the detection system, not the model.

Finding #4: Stability ≠ Accuracy

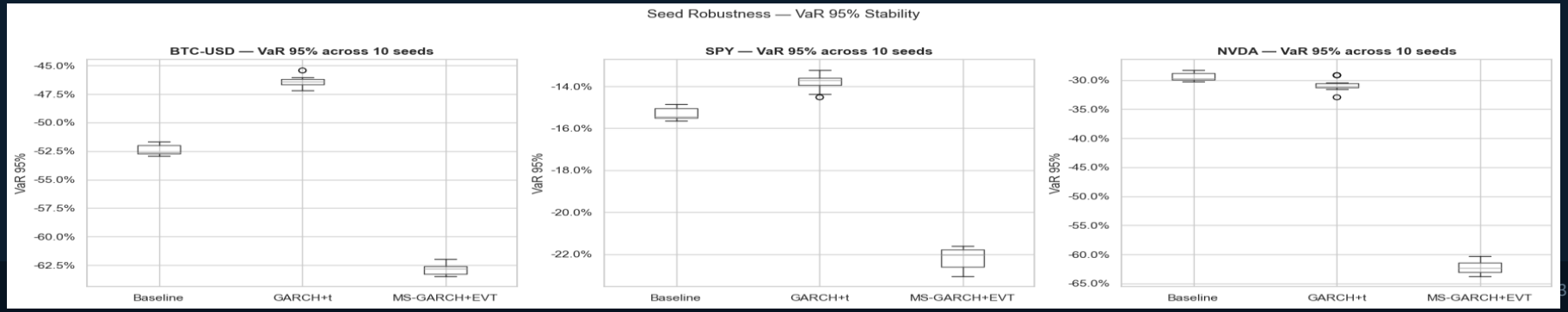
The original MS-GARCH (pre-fix) was the most stable model (lowest CV across seeds)

and also the most wrong.

Bug: fitted GARCH per-regime on only 44 crisis observations → α collapsed to 0 → the "regime-switching" model was actually constant-vol with a label.

The Fix (unified spec):

1. Global GARCH: single α, β on the full series (persistence ~0.96 across all assets)
2. Per-regime ω: via variance targeting — each regime gets its own long-run variance
3. Per-regime GPD: tail fitted on regime-filtered exceedances (fallback to Normal if sparse)



Conclusions

No single universally superior model

Rankings change with horizon and asset. The app recommends the tier whose failure mode is least damaging for each configuration.

Stability does not imply accuracy

A model that converges consistently may converge to the wrong number. Only the backtest distinguishes stability from calibration.

MS-GARCH+EVT: best calibrated, not perfect

4.1% breach vs 5% expected. Passes Kupiec + Christoffersen. But 30% breach in crisis due to HMM detection lag.

ML audits, it does not replace

XGBoost acts as a nonparametric auditor. It validates that parametric models aren't hallucinating. Its role is oversight, not prediction.

Questions?

"Each tier fails differently.

The best model depends on which failure you can tolerate."

Tech stack: Python · Streamlit · GARCH · HMM · EVT/GPD · XGBoost · Monte Carlo

271 tests · 5 notebooks · 3 tiers · Kupiec + Christoffersen backtesting

streamlit run app/streamlit_app.py